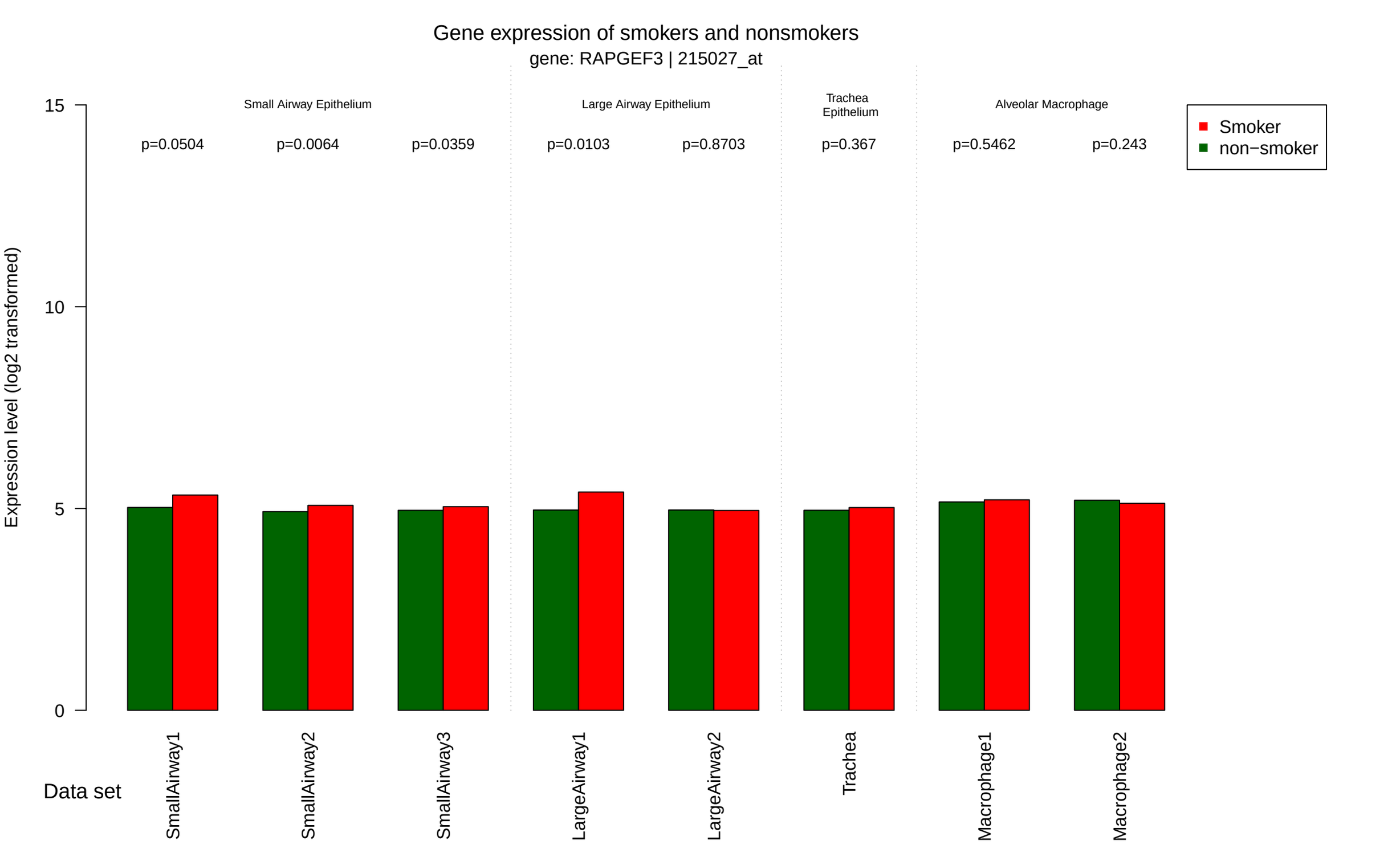
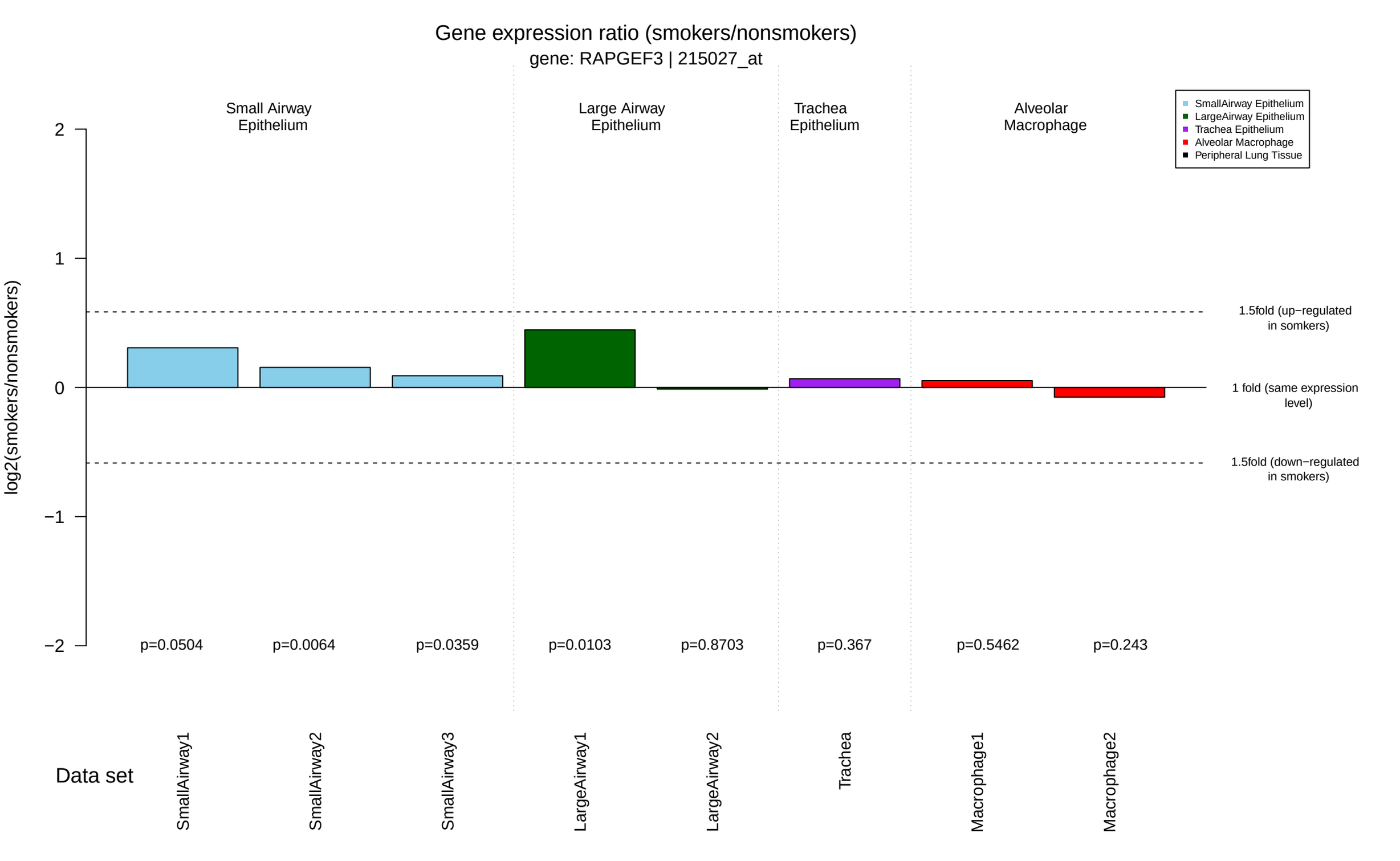
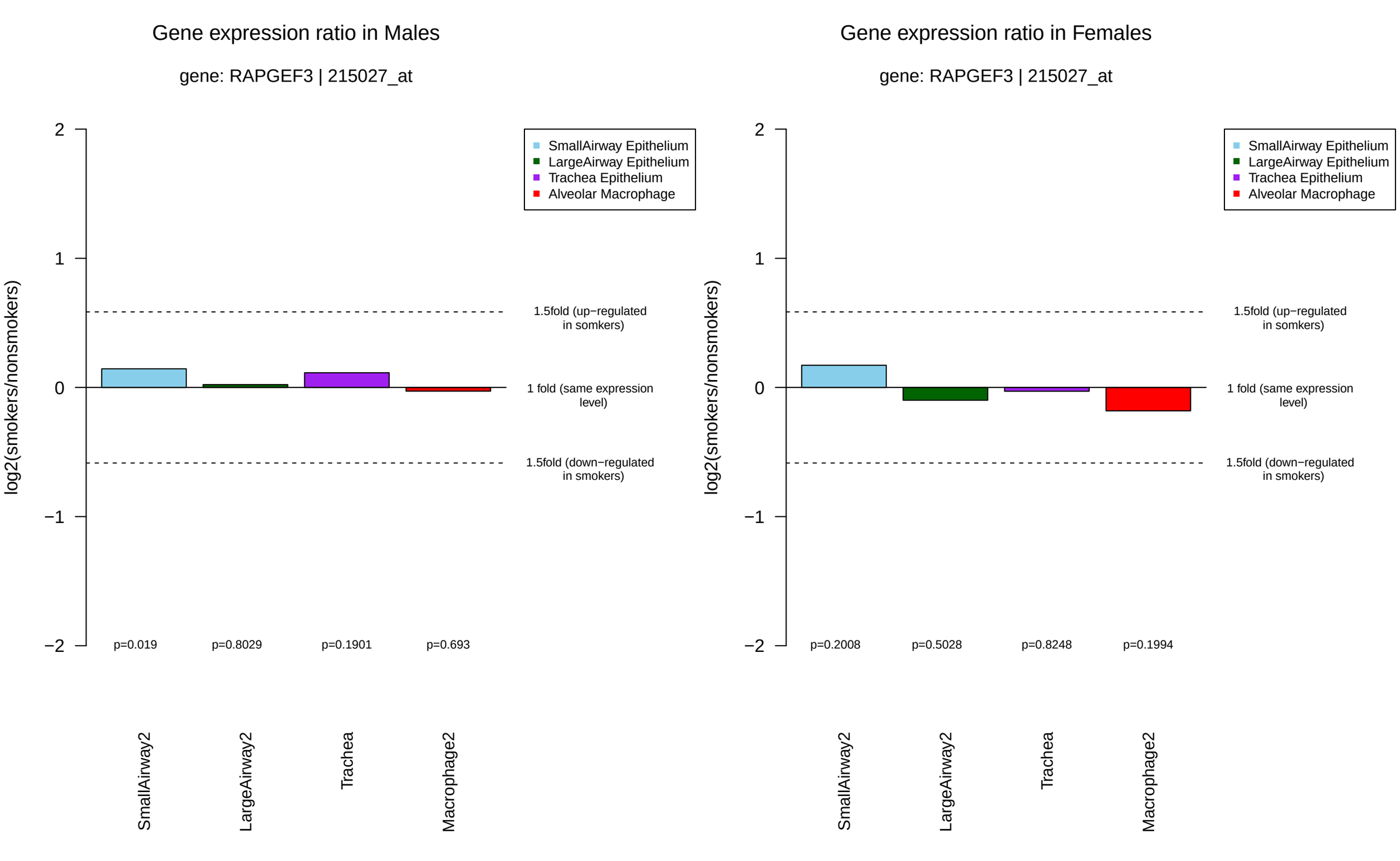
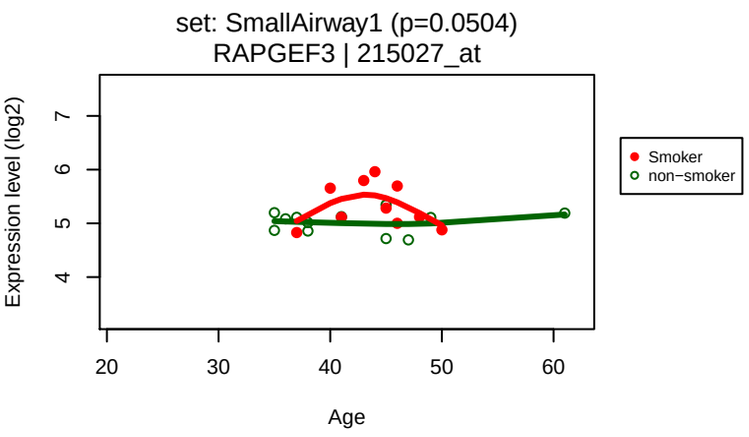
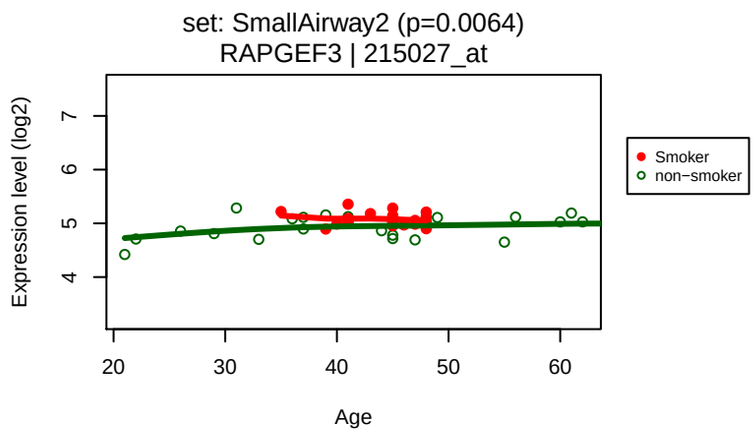
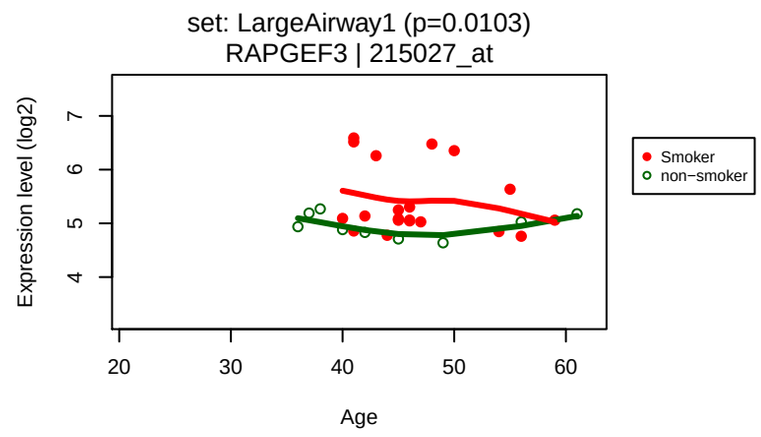

Gene expression of smokers and nonsmokers
gene: RAPGEF3 | 215027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0504
p=0.0064
p=0.0359
p=0.0103
p=0.8703
p=0.367
p=0.5462
p=0.243
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

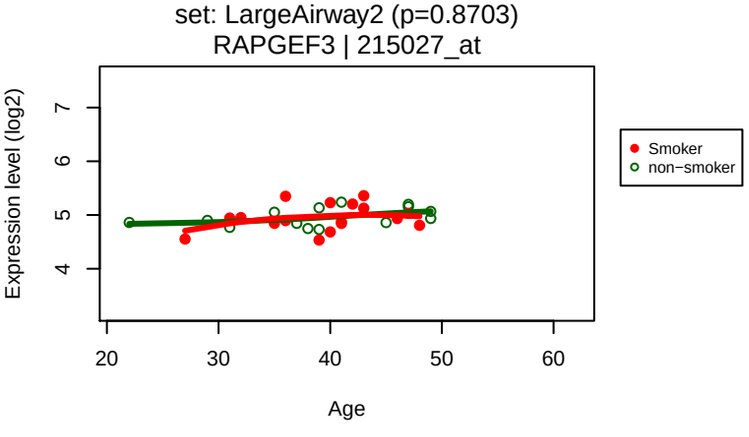

Gene expression ratio (smokers/nonsmokers)
gene: RAPGEF3 | 215027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0504
p=0.0064
p=0.0359
p=0.0103
p=0.8703
p=0.367
p=0.5462
p=0.243
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

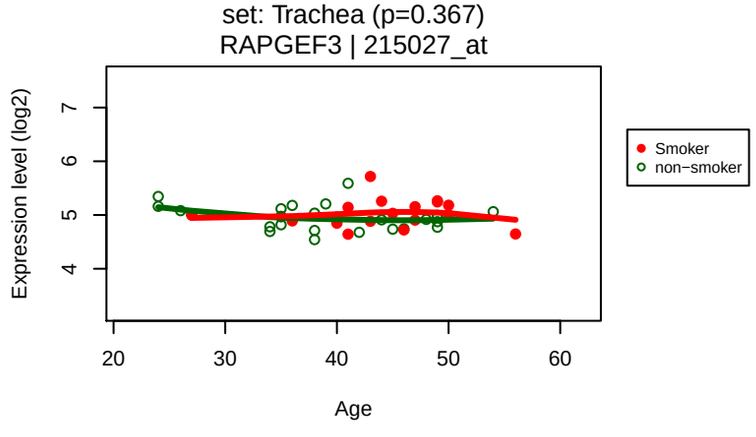

Gene expression ratio in Males
gene: RAPGEF3 | 215027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.019
p=0.8029
p=0.1901
p=0.693
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPGEF3 | 215027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2008
p=0.5028
p=0.8248
p=0.1994
SmallAirway2
LargeAirway2
Trachea
Macrophage2

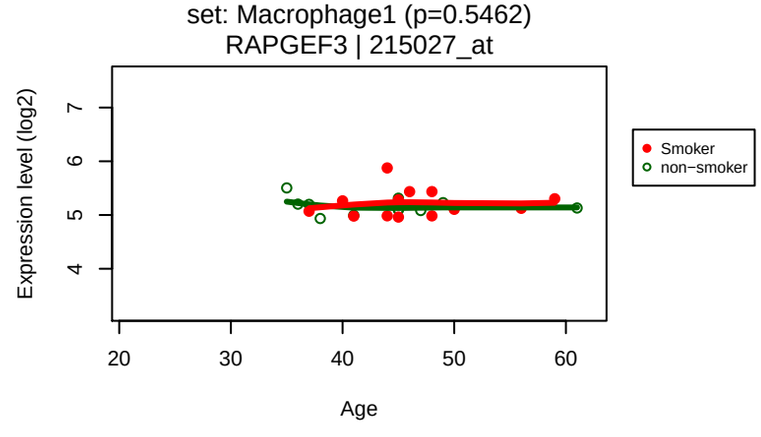

set: SmallAirway1 (p=0.0504)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker

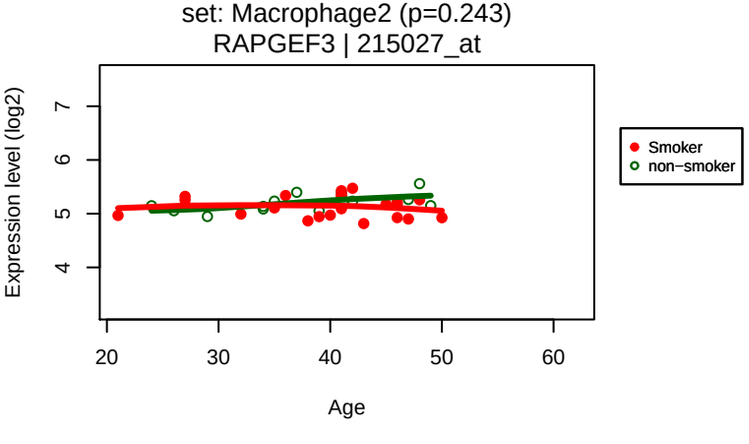

set: SmallAirway2 (p=0.0064)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker

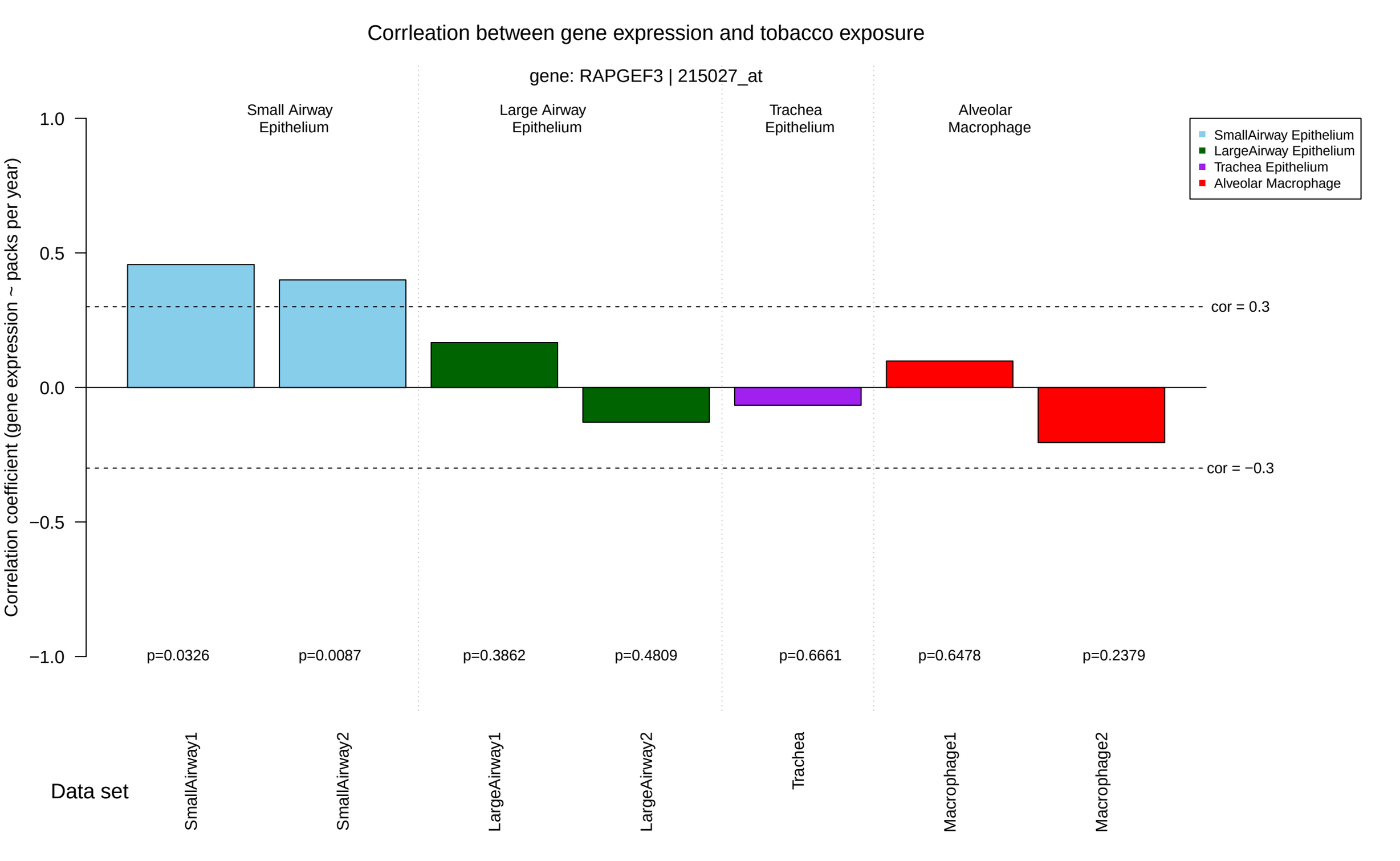

set: LargeAirway1 (p=0.0103)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8703)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.367)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5462)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.243)
RAPGEF3 | 215027_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPGEF3 | 215027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0326
p=0.0087
p=0.3862
p=0.4809
p=0.6661
p=0.6478
p=0.2379
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2